Reduction to difficult graphs via edge-disjoint barbells

This note rewrites the graph-theoretic reduction (Section 1.4 and Section 2 of the paper) along the plan in Section 0. Section 1 summarizes the chapter, and Section 2 gives the statements and complete proofs. The note is self-contained except for the final combination with Theorem 1.12 of the paper (the analysis of difficult graphs), which is not changed.

0 The plan (message of the advisor)

We can make the statement from cycles to barbells a purely graph-theoretic statement: there is no need to assume that the cycles are short. It should just be a statement that a connected subgraph with k edge-disjoint cycles has a large number of barbells.

The next step is an easy analysis of how many edges the whole graph has, and the conclusion that we must have enough short barbells, which leads to an exponentially small probability of SAT.

Here is the plan for Step 1 of the proof:

a) We eliminate all vertices of degree 0 and 1, (i) improving the excess, (ii) not changing the probability of SAT, and (iii) keeping the number of edges in G large. We then remove all connected components with excess ≤ 0. Again, the number of edges in the graph must remain large.

b) We reduce the largest degree $\Delta(G)$ to a constant Δ_0 (which depends on the constant in the linear excess): we (i) may only increase the probability of SAT, (ii) add vertices and edges, and (iii) decrease the excess, but still keep it positive and linear in n.

c) We state that graphs with many short edge-disjoint cycles are easy to analyze, and show that $\Pr[\text{SAT}] < e^{-\Omega(n)}$. We first prove/state a lemma. Lemma (graph theory): if a connected graph has k edge-disjoint cycles, then it has a large number of barbells with endpoints at those cycles.

   Then we state the corollary: if a graph has many (linear in n) edge-disjoint short cycles and positive excess in every connected component, then $\Pr[\text{SAT}] < e^{-\Omega(n)}$. The proof of this corollary is easy: we look at all connected components and consider two cases: (i) only one short cycle in the component; (ii) ≥ 2 short cycles in the component. In case (ii) we consider a minimal spanning subgraph connecting all those edge-disjoint cycles in the component and apply the graph-theoretic lemma. In case (i) we just find one barbell using only vertices of the connected component, which are disjoint from all remaining vertices. We then do a simple counting argument for the total number of edges in the graph and conclude that many of the barbells cannot have very long paths (otherwise there would be too many edges).

d) Finally, we can say that the graph can have only a sublinear number of loops (they form natural edge-disjoint short cycles) and a sublinear number of vertices with non-tree-like neighborhoods. Here the only claim to be stated and proved is that in bounded-degree graphs ($\Delta(G) < \Delta_0$), if there are ρn vertices with non-tree-like neighborhoods of constant radius R, then there is a constant $C(\Delta_0, R) > 0$ such that G has $C\rho n$ edge-disjoint cycles. The proof of this claim is straightforward: consider all vertices with non-tree-like neighborhoods, and notice that each such cycle may intersect at most a constant number of other such cycles.

1 Summary of the chapter

This section states all results of the chapter in the form in which they are proved in Section 2; each statement is repeated there word for word, with its proof. The notation is that of the paper and is recalled in Section 2.1. In particular the degree bound Δ_0 of the plan is the constant Δ of the paper, $c(\cdot)$ always counts components, and (H) is the hypothesis: no self-loops, all degrees in $[2, \Delta]$, and $\xi(H) \geq \delta|V(H)|$. A graph is *difficult* if it satisfies (H) and is (R, ρ)-locally tree-like.

1.1 The main result

Theorem 2.15 (Reduction to difficult graphs)**.** *Fix $\varepsilon > 0$, and let Δ, δ be as in Proposition 2.3. For every integer $R \geq 1$ and every $\rho \in (0, 1]$ there is $\beta = \beta(\varepsilon, R, \rho) > 0$ such that for every multigraph G with n vertices and $\xi(G) \geq \varepsilon n$, one of the following holds:*

(a) $\Pr(\Phi(G)\ SAT) \leq e^{-\beta n}$;

(b) there is a multigraph H that satisfies (H) *with Δ, δ, is (R, ρ)-locally tree-like, has $|V(H)| \geq \beta n$, and satisfies* $\Pr(\Phi(G)\ SAT) \leq \Pr(\Phi(H)\ SAT)$.

Combined with the paper, case (b) also gives $\Pr(\Phi(G)\ \mathrm{SAT}) \leq e^{-\Omega(n)}$. The paper's Theorem 1.12 gives $R_0(\Delta, \delta)$ and $\rho_1(\Delta, \delta)$; apply the theorem above with $R = R_0$ and $\rho = \rho_1$. Then H is difficult. Since $c(\mathrm{Sol}(\Phi(H))) \geq 1$ whenever $\Phi(H)$ is satisfiable, the main identity (Theorem 1.10) together with Theorem 1.12 gives $\Pr(\Phi(H)\ \mathrm{SAT}) \leq \mathbb{E}\, c(\mathrm{Sol}(\Phi(H))) \leq 2e^{-\gamma|V(H)|/2}$, with $|V(H)| \geq \beta n$. There is no circularity, because R_0 and ρ_1 depend only on Δ and δ.

1.2 Step 1: cleaning (plan (a) and (b))

Lemma 2.1 (Deleting low-degree vertices)**.** *Deleting an isolated vertex, or a vertex of degree 1 together with its edge, does not change satisfiability and increases the excess by $3/2$, respectively $1/2$.*

Lemma 2.2 (Splitting vertices)**.** *Fix an integer $\Delta \geq 4$. Every finite multigraph G with m edges can be changed into a finite multigraph H with maximum degree at most Δ and with*

$$N := |V(H)| - |V(G)| = |E(H)| - |E(G)| \leq \frac{2m}{\Delta - 2},$$

in which no vertex has degree 0 or 1 unless it had this degree in G. Moreover, there is a coupling of the random signs under which $\{\Phi(G)\ SAT\} \subseteq \{\Phi(H)\ SAT\}$, and under which the signs on H are independent and uniform, so that they have the law of $\Phi(H)$.

Proposition 2.3 (Cleaning)**.** *Fix $\varepsilon > 0$, and let*

$$\Delta := 2 + \left\lceil \max\left\{4, \frac{6}{\varepsilon}\right\} \right\rceil, \qquad \delta := \frac{\varepsilon}{2(4+\varepsilon)}.$$

For every multigraph G with n vertices and $\xi(G) \geq \varepsilon n$ there is a multigraph H_1 such that

(i) $2 \leq \deg_{H_1}(v) \leq \Delta$ *for every vertex v;*

(ii) every component of H_1 has positive excess;

(iii) $\xi(H_1) \geq 2\delta|V(H_1)|$;

(iv) $|V(H_1)| \geq (\varepsilon/\Delta)\, n$;

(v) $\Pr(\Phi(G)\ SAT) \leq \Pr(\Phi(H_1)\ SAT)$.

The graph H_1 is obtained by splitting first, then deleting vertices of degree $0, 1$ (Lemma 2.1), and then deleting components of nonpositive excess.

1.3 Step 2: many edge-disjoint cycles (plan (c))

Definition 2.5 (Barbell)**.** A *barbell* is a triple $Y = (D_1, D_2, P)$, where D_1, D_2 are edge-disjoint cycles, its *ends*, and P is a path from a vertex of D_1 to a vertex of D_2 that uses no edge of D_1 or D_2. Its edge set is $E(Y) = E(D_1) \cup E(D_2) \cup E(P)$. If D_1 and D_2 share a vertex, P may be this single vertex.

Here a cycle may be a self-loop or a pair of parallel edges.

Lemma 2.4 (Forcing cycles)**.** *Let C be a cycle and u a vertex of C. There is a sign pattern on $E(C)$ for which every satisfying assignment of the clauses on $E(C)$ has $x_u = 1$, and one for which every satisfying assignment has $x_u = 0$.*

Lemma 2.6 (Barbells give contradictions)**.** *Every barbell supports an unsatisfiable 2-CNF. Hence, if a barbell Y is a subgraph of G, then with probability at least $4^{-|E(Y)|}$ the signs on $E(Y)$ make $\Phi(G)$ unsatisfiable.*

This is the letter Q: one cycle forces the value 1 at the start of the path (Lemma 2.4), the clauses on the path carry it to the other cycle, and that cycle forces the value 0 there.

Lemma 2.7 (From walks to barbells)**.** *Let D_1, D_2 be edge-disjoint cycles, and let W be a walk from a vertex of D_1 to a vertex of D_2 that uses no edge of D_1 or D_2. Then there is a path P with $E(P) \subseteq E(W)$ such that (D_1, D_2, P) is a barbell.*

Lemma 2.8 (Positive excess gives two cycles)**.** *Every finite connected multigraph G with $\xi(G) > 0$ contains two edge-disjoint cycles.*

Corollary 2.9 (Positive excess gives a barbell)**.** *Every finite connected multigraph G with $\xi(G) > 0$ contains a barbell.*

The corollary follows by joining the two cycles of Lemma 2.8 by a shortest path between them. It handles the components that contain only one of the given cycles, in particular graphs with many components, each with a single short cycle. The hypothesis is positive excess, not positive cycle rank: a theta graph has $\nu = 2$ and no two edge-disjoint cycles.

Theorem 2.10 (Barbell lemma)**.** *Let G be a connected multigraph that contains pairwise edge-disjoint cycles $C_1, \dots, C_k$. Then G contains at least $\lceil (k-1)/3 \rceil$ pairwise edge-disjoint barbells, all of whose ends are among $C_1, \dots, C_k$.*

This is the purely graph-theoretic statement asked for in the plan: no length condition on the cycles. The proof is a bottom-up pairing on a spanning tree in which each cycle is contracted to a node. Every subtree passes at most one unmatched cycle upward, and ordinary nodes pair the arriving cycles. A cycle node with zero arrivals passes its own cycle up; with one arrival it pairs the arrival with its own cycle; with at least two arrivals it pairs them through edge-disjoint arcs of its cycle and discards its own cycle. At the end every cycle is an end of one of the b barbells (at most $2b$ cycles), the one cycle left unmatched at the root (at most 1), or discarded. Every discarded cycle is charged to a barbell created at the same node, so at most b cycles are discarded. Hence $k \le 2b + 1 + b$. The bound is sharp: join a vertex by an edge to each of t gadgets, each gadget being a triangle (entered at one vertex) with one more triangle hanging at each of its two other vertices. This gives $k = 3t$ edge-disjoint triangles. At most two triangles of each gadget can be ends: a barbell with the middle triangle as an end uses all of its edges, and an outer triangle that is not its partner can reach any other cycle only through an edge of the middle triangle. So at most $2t$ triangles are ends, and there are at most t edge-disjoint barbells.

Corollary 2.12 (Many edge-disjoint cycles)**.** *Let G be a multigraph with n vertices and maximum degree at most Δ in which every component has positive excess. If G contains at least αn pairwise edge-disjoint cycles, where $\alpha > 0$, then*

$$\Pr(\Phi(G) \text{ } SAT) \le \exp\Big(-\frac{\alpha}{8}\, 4^{-4\Delta/\alpha}\, n\Big).$$

Proof idea: let a component contain k_j of the cycles. If $k_j \geq 2$, Theorem 2.10 gives $\lceil (k_j-1)/3 \rceil \geq k_j/4$ edge-disjoint barbells; the inequality is $1 \geq k_j/4$ for $k_j = 2, 3$, and $4(k_j-1) \geq 3k_j$ for $k_j \geq 4$. If $k_j = 1$, Corollary 2.9 gives one barbell, and $1 \geq k_j/4$. Summing over components gives at least $\alpha n/4$ edge-disjoint barbells. The constant $1/4$ is the best uniform one, since $k_j = 4$ only guarantees one barbell. Since $|E(G)| \leq \Delta n/2$, at least $\alpha n/8$ of these barbells have at most $4\Delta/\alpha$ edges, and they fail independently.

1.4 Step 3: few cycles (plan (d))

Lemma 2.13 (Non-tree balls give disjoint cycles)**.** *Let G have maximum degree at most Δ, let $R \geq 1$, and suppose that N vertices v have a ball $B_R(v)$ that is not a tree. Then G contains at least N/M_{2R} pairwise vertex-disjoint cycles, each of length at most $2R+1$.*

This is the claim of plan (d), with $C(\Delta_0, R) = 1/M_{2R}$.

Lemma 2.14 (Removing self-loops)**.** *Fix Δ, $\delta > 0$, $R \geq 1$ and $\rho \in (0, 1]$. Let G have n vertices, $2 \leq \deg_G(v) \leq \Delta$ for all v, and $\xi(G) \geq 2\delta n$; suppose that G has at most δn self-loops and at most $(2\delta\rho/\Delta)n$ vertices with a non-tree ball $B_R(v)$. Delete all self-loops, then repeatedly delete vertices of degree 0 or 1, and call the result H. Then H satisfies* (H) *with Δ, δ, it is (R, ρ)-locally tree-like, $|V(H)| \geq (2\delta/\Delta)n$, and $\Pr(\Phi(G)\ SAT) \leq \Pr(\Phi(H)\ SAT)$.*

1.5 How the statements give the main result

Theorem 2.15 has two possible conclusions: (a) the satisfiability probability is exponentially small, or (b) it is bounded by that of a difficult graph H with $|V(H)| \geq \beta n$. Which one we get depends on the number of edge-disjoint cycles of the cleaned graph.

Let H_1 be the graph of Proposition 2.3, $n_1 = |V(H_1)| \geq (\varepsilon/\Delta)n$, and

$$\alpha := \frac{2\delta\rho}{\Delta M_{2R}} \leq \delta.$$

- *H_1 has at least αn_1 edge-disjoint cycles.* Then Corollary 2.12 gives $\Pr(\Phi(H_1)\ \mathrm{SAT}) \leq e^{-\Omega(n_1)}$. Since $\Pr(\Phi(G)\ \mathrm{SAT}) \leq \Pr(\Phi(H_1)\ \mathrm{SAT})$ and $n_1 \geq (\varepsilon/\Delta)n$, this is conclusion (a).
- *H_1 has fewer than αn_1 edge-disjoint cycles.* Self-loops are edge-disjoint cycles, so H_1 has fewer than $\alpha n_1 \leq \delta n_1$ self-loops. By Lemma 2.13 it has fewer than $\alpha M_{2R} n_1 = (2\delta\rho/\Delta)n_1$ vertices with a non-tree ball. So Lemma 2.14 applies to H_1, and the resulting H is difficult, with $|V(H)| \geq (2\delta/\Delta)n_1 \geq (2\delta\varepsilon/\Delta^2)n$. This is conclusion (b).

The single threshold α is chosen for two reasons. First, $\alpha \leq \delta$, so the fewer than δn_1 self-loops cost at most half of the excess $\xi(H_1) \geq 2\delta n_1$, and $\xi(H) \geq \delta n_1$ survives. Second, $\alpha M_{2R} = 2\delta\rho/\Delta$: the graph H may have fewer vertices than H_1, but $|V(H)| \geq (2\delta/\Delta)n_1$, so the fewer than $(2\delta\rho/\Delta)n_1$ non-tree balls are still at most $\rho|V(H)|$. The constants are

$$\beta = \min\Big\{ \frac{\varepsilon}{\Delta} \cdot \frac{\alpha}{8}\, 4^{-4\Delta/\alpha},\ \frac{2\delta\varepsilon}{\Delta^2} \Big\}.$$

1.6 Changes with respect to the plan

(1) *Order of steps a) and b) of the plan.* Splitting a vertex lowers the excess of its component, so a component of small positive excess may become nonpositive. We therefore split first and clean afterwards. Splitting creates no vertex of degree 0 or 1, so the cleaning is done only once.

(2) *No dense case.* The paper uses the first-moment bound for $m \geq 2.41n$ only to control the number of vertices added by the splitting. A direct computation (Proposition 2.3) shows that the splitting works for every density, so the first-moment bound is no longer needed. The splitting adds at most $2m/(\Delta-2)$ vertices and lowers the excess by at most $m/(\Delta-2)$, where $m = \frac{3}{2}n + \xi(G)$. Both the n-part and the $\xi(G)$-part of this loss are small fractions of $\xi(G) \geq \varepsilon n$ once $\Delta \geq 2 + 6/\varepsilon$, however large m is. What has to stay linear is the excess *relative to the new number of vertices*, because (H) asks for $\xi(H) \geq \delta|V(H)|$.

(3) *"Short" is not needed in Step 2.* Corollary 2.12 assumes only linearly many edge-disjoint cycles. Bounded size of the barbells follows from the edge count $|E| \leq \Delta|V|/2$.

(4) *Case (i) of step c) needs a lemma.* A component with only one of the chosen cycles still contains a barbell, but this uses its positive excess: Lemma 2.8 shows that a connected graph of positive excess contains two edge-disjoint cycles. This is where graphs with many components, each with a single short cycle, are handled. The minimal spanning subgraph in case (ii) is not needed, since the barbell lemma holds for every connected graph.

(5) *"Sublinear" becomes quantitative.* Step 3 needs at most αn_1 self-loops and non-tree balls, with the explicit constant α above; a genuine $o(n)$ bound is neither available nor needed.

(6) *After deleting the self-loops* one must delete the new vertices of degree $0, 1$, because (H) asks for minimum degree 2. Components of nonpositive excess need not be deleted again: (H) does not ask for positive excess per component.

1.7 What disappears from the paper

The following parts of the paper are no longer needed:

- the first-moment bound (Lemma 1.3) as a case of the proof; it can remain as motivation;
- the cycle-rank criterion for unsatisfiable support (Lemma 1.6, proved by Lemmas 2.3 and 2.4), and with it the reference to Karve–Hirani;
- the separate treatment of self-loops (Lemma 1.7 and Lemma 2.5);
- the local-tree counting lemma (Lemma 1.8), with its radii s, $R_1 = 2s + R$ and three cases;
- the packing of disjoint balls (Corollary 1.9).

Lemmas 1.4 and 1.5 (bounded degree and positive-excess components) are replaced by Proposition 2.3. The removed parts are replaced by one graph-theoretic theorem (the barbell lemma) and one counting argument (Corollary 2.12).

2 Implementation

2.1 Setting

A *multigraph* may have parallel edges and self-loops. Each vertex v carries a Boolean variable x_v, and each edge uv carries one of the four clauses $x_u \vee x_v$, $x_u \vee \neg x_v$, $\neg x_u \vee x_v$, $\neg x_u \vee \neg x_v$: each end of the edge gets a sign, which decides whether its variable appears negated. The two signs of an edge are independent and uniform, and different edges are independent. A self-loop at v carries a clause on x_v alone, again with two independent uniform signs. The resulting random 2-CNF is $\Phi(G)$. Deleting an edge deletes its clause, which can only make the formula easier to satisfy.

For a multigraph H we write $\xi(H) = |E(H)| - \frac{3}{2}|V(H)|$ for its *excess*; it is additive over components. In the degree $\deg_H(v)$ a self-loop at v adds 2. The ball $B_R(v)$ is the subgraph induced by the vertices u with $\mathrm{dist}(u, v) \leq R$; it is connected. If the maximum degree is at most

Δ, then $|V(B_R(v))| \le M_R := 1 + \Delta \sum_{j=0}^{R-1} (\Delta - 1)^j$. A graph is (R, ρ)-*locally tree-like* if at most $\rho|V|$ vertices v have a ball $B_R(v)$ that is not a tree.

A *cycle* of length $L \ge 1$ is a self-loop ($L = 1$), two parallel edges ($L = 2$), or, for $L \ge 3$, a closed path through L distinct vertices. A *path* may consist of a single vertex. A multigraph H *supports an unsatisfiable 2-CNF* if for some sign pattern on $E(H)$ the clauses on $E(H)$ are unsatisfiable.

The goal of the reduction is the following hypothesis, for constants Δ and $\delta > 0$:

(H) H has no self-loops, $2 \le \deg_H(v) \le \Delta$ for every vertex, and $\xi(H) \ge \delta|V(H)|$.

A graph that satisfies (H) and is (R, ρ)-locally tree-like is *difficult*.

2.2 Step 1: bounded degree and positive excess

Lemma 2.1 (Deleting low-degree vertices)**.** *Deleting an isolated vertex, or a vertex of degree* 1 *together with its edge, does not change satisfiability and increases the excess by* $3/2$*, respectively* $1/2$.

Proof. An isolated variable does not occur in the formula. A variable that occurs in one clause only can always be chosen to satisfy it. The changes of the excess are $\frac{3}{2}$ and $-1 + \frac{3}{2} = \frac{1}{2}$. □

Lemma 2.2 (Splitting vertices)**.** *Fix an integer* $\Delta \ge 4$*. Every finite multigraph* G *with* m *edges can be changed into a finite multigraph* H *with maximum degree at most* Δ *and with*

$$N := |V(H)| - |V(G)| = |E(H)| - |E(G)| \le \frac{2m}{\Delta - 2},$$

in which no vertex has degree 0 *or* 1 *unless it had this degree in* G*. Moreover, there is a coupling of the random signs under which* $\{\Phi(G)\ SAT\} \subseteq \{\Phi(H)\ SAT\}$*, and under which the signs on* H *are independent and uniform, so that they have the law of* $\Phi(H)$*.*

Proof. Let $b := \Delta - 2 \ge 2$. For every vertex v with $\deg_G(v) > b$, let $k_v = \lceil \deg_G(v)/b \rceil$, replace v by k_v copies, split the $\deg_G(v)$ edge ends at v into k_v nonempty groups of size at most b, give one group to each copy, and join the copies by a path. The number of added vertices equals the number of added path edges, namely $N = \sum_v (k_v - 1) \le \sum_v \deg_G(v)/b = 2m/(\Delta - 2)$. Every copy has at most b original edge ends and at most two path edges, so its degree is at most Δ, and since every group is nonempty and $k_v \ge 2$, every copy has degree at least 2.

For the coupling, fix a split vertex with copies $1, \dots, k_v$ along the path. First put on each path edge ij both clauses $(\neg x_i \vee x_j) \wedge (x_i \vee \neg x_j)$; they force all copies to be equal, so the formula is satisfiable exactly when $\Phi(G)$ is. Next choose independent fair bits g_i, one per copy, and replace x_i by $x_i \oplus g_i$ in every clause; this does not change satisfiability. Finally keep on every path edge only one of its two clauses, chosen independently and uniformly. The result is a subformula, so it is satisfiable whenever $\Phi(G)$ is.

It remains to show that the resulting signs are independent and uniform. Consider one path with copies $1, \dots, k_v$. Its randomness consists of the bits $g_1, \dots, g_{k_v}$ and the $k_v - 1$ choices of kept clauses, so $2^{k_v} \cdot 2^{k_v - 1}$ equally likely outcomes, and it produces the signs on the $k_v - 1$ path edges, $4^{k_v - 1}$ possible patterns. After the xor, the sign at end i of the edge $\{i, i+1\}$ is the sign of x_i in the kept clause flipped by g_i, and likewise at end $i + 1$. So for fixed g_i, the target sign at end i determines which clause is kept, and then the target sign at end $i + 1$ determines g_{i+1}. Hence, once g_1 is fixed, every target pattern arises from exactly one outcome, so every pattern arises from exactly two of the $2 \cdot 4^{k_v - 1}$ outcomes, and the signs on the path edges are uniform. Different paths use independent randomness. Finally, every sign at an end of an original edge is the original sign flipped by the bit of its copy. The original signs are uniform and independent of all bits and choices, so the flipped signs are again uniform and independent of them, and hence of the signs on the path edges. □

Proposition 2.3 (Cleaning). *Fix $\varepsilon > 0$, and let*

$$\Delta := 2 + \left\lceil \max\Big\{4, \frac{6}{\varepsilon}\Big\} \right\rceil, \qquad \delta := \frac{\varepsilon}{2(4+\varepsilon)}.$$

For every multigraph G with n vertices and $\xi(G) \geq \varepsilon n$ there is a multigraph H_1 such that

(i) $2 \leq \deg_{H_1}(v) \leq \Delta$ *for every vertex v;*

(ii) every component of H_1 has positive excess;

(iii) $\xi(H_1) \geq 2\delta |V(H_1)|$*;*

(iv) $|V(H_1)| \geq (\varepsilon/\Delta)\, n$*;*

(v) $\Pr(\Phi(G)\ SAT) \leq \Pr(\Phi(H_1)\ SAT)$.

Proof. Write $\xi = \xi(G)$ and $m = |E(G)| = \frac{3}{2}n + \xi$, and let $\theta := 1/(\Delta - 2)$, so that $\theta \leq \frac{1}{4}$ and $\theta \leq \varepsilon/6$.

Splitting. Lemma 2.2 gives H' with maximum degree at most Δ and $N \leq 2\theta m$ new vertices and edges. Each new vertex with its new edge lowers the excess by $\frac{1}{2}$, so

$$\xi(H') = \xi - \frac{N}{2} \geq \xi - \theta\Big(\frac{3}{2}n + \xi\Big) \geq \frac{3}{4}\xi - \frac{\varepsilon}{4}n \geq \frac{\xi}{2},$$

using $\theta \leq \frac{1}{4}$, $\frac{3}{2}\theta \leq \frac{\varepsilon}{4}$ and $\varepsilon n \leq \xi$. Also $|V(H')| = n + N \leq n + 2\theta(\frac{3}{2}n + \xi) \leq 2n + \frac{\xi}{2}$, hence

$$\frac{\xi(H')}{|V(H')|} \geq \frac{\xi/2}{2n + \xi/2} = \frac{\xi}{4n + \xi} \geq \frac{\varepsilon}{4+\varepsilon} = 2\delta,$$

since the middle expression increases with $\xi \geq \varepsilon n$.

Cleaning. Repeatedly delete vertices of degree 0 or 1, and then delete every component of nonpositive excess; call the result H_1. By Lemma 2.1 the first deletions do not lower the excess, and since the excess is additive over components, neither do the second. The number of vertices does not grow, so $\xi(H_1) \geq \xi(H') \geq 2\delta|V(H')| \geq 2\delta|V(H_1)|$. Deletions do not raise degrees, the first deletions stop only when all degrees are at least 2, and deleting whole components does not change the remaining degrees; this gives (i), and (ii) holds by construction. Since $\xi(H_1) \leq |E(H_1)| \leq \frac{\Delta}{2}|V(H_1)|$ and $\xi(H_1) \geq \xi(H') \geq \xi/2 \geq \varepsilon n/2$, we get (iv). Finally, (v) follows from the coupling of Lemma 2.2, from Lemma 2.1, and from the fact that deleting components deletes clauses. □

2.3 Local contradictions: the letter Q

Lemma 2.4 (Forcing cycles). *Let C be a cycle and u a vertex of C. There is a sign pattern on $E(C)$ for which every satisfying assignment of the clauses on $E(C)$ has $x_u = 1$, and one for which every satisfying assignment has $x_u = 0$.*

Proof. Write the cycle as $u = u_0, u_1, \ldots, u_L = u$, where L is its length and consecutive vertices are joined by the edges of C. Put $(x_{u_0} \vee x_{u_1})$ on the edge u_0u_1, and $(\neg x_{u_i} \vee x_{u_{i+1}})$ on the edge u_iu_{i+1} for $1 \leq i \leq L-1$. For $L = 1$ this is the single clause $(x_u \vee x_u)$. If $x_u = 0$, the first clause gives $x_{u_1} = 1$, and the others give $x_{u_2} = \cdots = x_{u_L} = 1$, which contradicts $x_{u_L} = x_u = 0$. Replacing x_u by $\neg x_u$ in these clauses gives a pattern that forces $x_u = 0$. □

Definition 2.5 (Barbell). A *barbell* is a triple $Y = (D_1, D_2, P)$, where D_1, D_2 are edge-disjoint cycles, its *ends*, and P is a path from a vertex of D_1 to a vertex of D_2 that uses no edge of D_1 or D_2. Its edge set is $E(Y) = E(D_1) \cup E(D_2) \cup E(P)$. If D_1 and D_2 share a vertex, P may be this single vertex.

Lemma 2.6 (Barbells give contradictions)**.** *Every barbell supports an unsatisfiable 2-CNF. Hence, if a barbell Y is a subgraph of G, then with probability at least $4^{-|E(Y)|}$ the signs on $E(Y)$ make $\Phi(G)$ unsatisfiable.*

Proof. Let $Y = (D_1, D_2, P)$ with $P = u_0u_1 \cdots u_k$, $u_0 \in V(D_1)$, $u_k \in V(D_2)$. The three edge sets are disjoint, so we may choose the signs on them separately. By Lemma 2.4, choose signs on D_1 that force $x_{u_0} = 1$ and signs on D_2 that force $x_{u_k} = 0$, and put $(\neg x_{u_i} \vee x_{u_{i+1}})$ on the edge u_iu_{i+1}. A satisfying assignment would have $x_{u_0} = 1$, hence $x_{u_1} = \cdots = x_{u_k} = 1$, which contradicts $x_{u_k} = 0$. (For $k = 0$ the two cycles force opposite values at the same vertex.) The cycle D_1 with the path P is the letter Q: its tail u_k is forced. The second claim holds because each edge has four sign patterns, and adding clauses keeps a formula unsatisfiable. □

Lemma 2.7 (From walks to barbells)**.** *Let D_1, D_2 be edge-disjoint cycles, and let W be a walk from a vertex of D_1 to a vertex of D_2 that uses no edge of D_1 or D_2. Then there is a path P with $E(P) \subseteq E(W)$ such that (D_1, D_2, P) is a barbell.*

Proof. Every walk contains a path between its endpoints all of whose edges are edges of the walk: repeatedly cut out a closed subwalk between two visits of the same vertex. □

Lemma 2.8 (Positive excess gives two cycles)**.** *Every finite connected multigraph G with $\xi(G) > 0$ contains two edge-disjoint cycles.*

Proof. Repeatedly delete a vertex of degree 1 with its edge, or replace a vertex w of degree 2 without self-loop and its edges wy, wz by one edge yz (a self-loop if $y = z$). Each step removes a vertex, so the process stops. Each step keeps the graph connected, raises the excess by $\frac{1}{2}$ (one vertex and one edge fewer), and maps edge-disjoint cycles back to edge-disjoint cycles: replace the edge yz by the two edges wy, wz. The final graph H has $\xi(H) > 0$; a vertex of degree 0, or of degree 2 with a self-loop, would be all of H and give $\xi(H) < 0$, so all degrees are at least 3. Let $n' = |V(H)|$, $m' = |E(H)| > \frac{3}{2}n'$, let D be a shortest cycle, of length g, and suppose that $H - E(D)$ is a forest. Then $m' - g \leq n' - 1$, so $g > \frac{n'}{2} + 1$. Hence $n' \geq 2$ (for $n' = 1$ we would have $g = 1$), so $g \geq 3$: H is simple, and D has no chord, since a chord would close a shorter cycle. Every vertex of D has degree at least 3 but only two neighbors on D, so it has a neighbor outside D. From $g > \frac{n'}{2} + 1$ we get $n' - g < g - 2$: the g vertices of D have their outside neighbors among fewer than g outside vertices, so by pigeonhole one outside vertex w has two neighbors $a \neq b$ on D. Then wa, wb and the shorter arc of D from a to b form a cycle of length at most $\lfloor g/2 \rfloor + 2 < g$ if $g \geq 5$. If $g \leq 4$, then $g + 1 \leq n' < 2g - 2$ (there is at least one outside vertex), which is impossible for $g = 3$; so $g = 4$, $n' = 5$, w is the only outside vertex, and w is adjacent to all four vertices of D, in particular to two consecutive ones, which gives a triangle. In every case this contradicts the choice of D.

Positive excess cannot be replaced by positive cycle rank, or even by $\nu(G) \geq 2$: in a theta graph (three internally disjoint paths between two vertices) every cycle is the union of two of the three paths, so any two cycles share a path. If the paths have lengths L_1, L_2, L_3, the theta graph has $L_1 + L_2 + L_3$ edges and $L_1 + L_2 + L_3 - 1$ vertices, so $\nu = 2$ but $\xi = \frac{3}{2} - \frac{1}{2}(L_1 + L_2 + L_3) \leq 0$. □

Corollary 2.9 (Positive excess gives a barbell)**.** *Every finite connected multigraph G with $\xi(G) > 0$ contains a barbell.*

Proof. Take edge-disjoint cycles D_1, D_2 from Lemma 2.8. If they share a vertex, we are done. Otherwise take a shortest path from $V(D_1)$ to $V(D_2)$. Its inner vertices lie on neither cycle and its endpoints lie on different cycles, so none of its edges joins two vertices of the same cycle; hence it uses no edge of D_1 or D_2, and Lemma 2.7 applies. □

2.4 The barbell lemma

Theorem 2.10 (Barbell lemma)**.** *Let G be a connected multigraph that contains pairwise edge-disjoint cycles $C_1, \dots, C_k$. Then G contains at least $\lceil (k-1)/3 \rceil$ pairwise edge-disjoint barbells, all of whose ends are among $C_1, \dots, C_k$.*

Proof. The auxiliary tree. Call an edge of G *free* if it lies on none of the C_i. Let Γ be the graph on $V(G) \cup \{w_1, \dots, w_k\}$ whose edges are the free edges of G that are not self-loops, and one edge $w_i v$ for every $v \in V(C_i)$. It is connected, since an edge xy of C_i is replaced by $x\, w_i\, y$. Fix a spanning tree $\mathcal{T}$ of Γ rooted at a vertex of G. A *cycle node* w_i has a parent $u_i \in V(C_i)$, and its children are distinct vertices of C_i other than u_i; a tree edge between two vertices of G is a free edge. Say that a subtree $\mathcal{T}_x$ *owns* its free tree edges and the edges of all C_i with $w_i \in \mathcal{T}_x$. Disjoint subtrees own disjoint edge sets, and $E(C_i)$ is owned by $\mathcal{T}_{w_i}$ but by no subtree of a child of w_i. The *top* of $\mathcal{T}_x$ is x if $x \in V(G)$, and u_i if $x = w_i$.

The algorithm. An *unmatched cycle at* a vertex v is a pair (C, W) of a cycle $C \in \{C_1, \dots, C_k\}$ and a walk W from a vertex of C to v that uses no edge of C. Processing $\mathcal{T}$ from the leaves up, every subtree $\mathcal{T}_x$ produces barbells and passes upward at most one unmatched cycle at its top; all of them use only edges owned by $\mathcal{T}_x$ and are pairwise edge-disjoint. The unmatched cycle passed up by a child y of x *arrives* at x: if $x, y \in V(G)$, its walk is extended by the free tree edge yx, so that it ends at x; if $y = w_i$, it already ends at $u_i = x$; if $x = w_i$, it ends at $y \in V(C_i)$. Arrivals from different children use disjoint owned edges (plus distinct free tree edges), so the walk of one arrival avoids the cycle of any other; and at $x = w_i$ no arrival uses an edge of C_i. By Lemma 2.7, every walk below that joins two of the cycles and avoids both of their edges contains a barbell.

- $x \in V(G)$. Pair the arrivals arbitrarily. For a pair $(C, W), (C', W')$, the walk $W\overline{W'}$ (the bar denotes reversal) joins C to C' and avoids both, so it gives a barbell. If one arrival is left over, pass it up.

- *$x = w_i$, with $C = C_i$, $u = u_i$, and arrivals at distinct vertices $a_1, \dots, a_r$ of C other than u.* If $r = 0$, pass up (C, u), with the walk of length 0. If $r = 1$, the walk of the arrival joins its cycle to C, which gives a barbell with ends $C^{(1)}, C$. If $r \geq 2$, then C has the three distinct vertices u, a_1, a_2, so it is a closed path. Number the a_j in the order in which they are met when going around C from u, put $a_{r+1} := u$, and let A_j be the arc of C from a_j to a_{j+1}; the arcs are pairwise edge-disjoint. For $t \leq r/2$, the walk $W^{(2t-1)} A_{2t-1} \overline{W^{(2t)}}$ joins $C^{(2t-1)}$ to $C^{(2t)}$ and avoids both, since $A_{2t-1} \subseteq C$; it gives a barbell. If r is odd, pass up $(C^{(r)}, W^{(r)} A_r)$, which ends at u. The cycle C is *discarded*.

The objects created at x use the edges of distinct arrivals, together with either all of $E(C_i)$ in a single object ($r \leq 1$) or distinct arcs of C_i ($r \geq 2$). Hence edge-disjointness and ownership are preserved.

Counting. Every cycle C_i is, at the end, an end of a barbell, the unmatched cycle passed up by the root, or discarded. A discarded cycle is charged to a barbell created at the same node, and such a barbell exists because $r \geq 2$. Different barbells are edge-disjoint and hence have different ends. With b barbells we get $k \leq 2b + 1 + b$, that is, $b \geq \lceil (k-1)/3 \rceil$. □

Remark 2.11 (Sharpness). The bound is attained. Join a vertex y by an edge yz_i to each of t gadgets, where gadget i is a triangle $Z_i = z_i a_i b_i$ with one more triangle T_i attached at a_i and one T_i' attached at b_i. There are $k = 3t$ edge-disjoint triangles. Suppose that all three triangles of gadget i are ends of pairwise edge-disjoint barbells. The barbell with end Z_i contains all of $E(Z_i)$, and its other end is not both T_i and T_i'; say it is not T_i'. The barbell with end T_i' is then a different barbell, so it uses no edge of Z_i. But its other end is a cycle other than T_i', and every such cycle can be reached from T_i' only through an edge of Z_i: removing $E(Z_i)$ separates T_i' from the rest of the graph. This is a contradiction. So at most $2t$ triangles are ends, and there are at

most $t = \lceil (k-1)/3 \rceil$ barbells. For the reduction any bound that is linear in k for $k \geq 2$ would do; we use $\lceil (k-1)/3 \rceil \geq k/4$ for $k \geq 2$.

2.5 Step 2: many edge-disjoint cycles

Corollary 2.12 (Many edge-disjoint cycles)**.** *Let G be a multigraph with n vertices and maximum degree at most Δ in which every component has positive excess. If G contains at least αn pairwise edge-disjoint cycles, where $\alpha > 0$, then*

$$\Pr(\Phi(G) \text{ SAT}) \leq \exp\Big(-\frac{\alpha}{8}\, 4^{-4\Delta/\alpha}\, n\Big).$$

Proof. Fix the cycles. Every cycle is connected and lies in one component; let k_j be the number of fixed cycles in the j-th component. In a component with $k_j \geq 2$, Theorem 2.10 gives $\lceil (k_j - 1)/3 \rceil \geq k_j/4$ edge-disjoint barbells. In a component with $k_j = 1$, Corollary 2.9 gives one barbell, and $1 \geq k_j/4$. Barbells in different components are edge-disjoint, so we have $b \geq \sum_j k_j/4 \geq \alpha n/4$ pairwise edge-disjoint barbells.

They are edge-disjoint, so together they use at most $|E(G)| \leq \Delta n/2$ edges. Hence fewer than $(\Delta n/2)/(4\Delta/\alpha) = \alpha n/8$ of them have more than $4\Delta/\alpha$ edges, and at least $\alpha n/4 - \alpha n/8 = \alpha n/8$ barbells have at most $4\Delta/\alpha$ edges. By Lemma 2.6, the signs on each of them make the formula unsatisfiable with probability at least $q := 4^{-4\Delta/\alpha}$, and these events are independent because the barbells are edge-disjoint. Therefore $\Pr(\Phi(G) \text{ SAT}) \leq (1-q)^{\alpha n/8} \leq e^{-q\alpha n/8}$. □

2.6 Step 3: few cycles

Lemma 2.13 (Non-tree balls give disjoint cycles)**.** *Let G have maximum degree at most Δ, let $R \geq 1$, and suppose that N vertices v have a ball $B_R(v)$ that is not a tree. Then G contains at least N/M_{2R} pairwise vertex-disjoint cycles, each of length at most $2R+1$.*

Proof. Let $B = B_R(v)$ be a ball that is not a tree, and let $\mathcal{T}$ be a breadth-first search tree of B from v; all its vertices have depth at most R, because a shortest path from v stays inside B. Since B is not a tree, it has an edge $e = xy$ outside $\mathcal{T}$, possibly a self-loop or an edge parallel to a tree edge. The edge e together with the tree paths from x and y to their last common ancestor is a cycle in B of length at most $2R+1$.

Choose centers greedily among the N vertices, removing every candidate at distance at most $2R$ from a chosen center. Each choice removes at most M_{2R} candidates, so at least N/M_{2R} centers are chosen. Their balls of radius R are pairwise vertex-disjoint, and so are the cycles in them. □

Lemma 2.14 (Removing self-loops)**.** *Fix Δ, $\delta > 0$, $R \geq 1$ and $\rho \in (0, 1]$. Let G have n vertices, $2 \leq \deg_G(v) \leq \Delta$ for all v, and $\xi(G) \geq 2\delta n$; suppose that G has at most δn self-loops and at most $(2\delta\rho/\Delta)n$ vertices with a non-tree ball $B_R(v)$. Delete all self-loops, then repeatedly delete vertices of degree 0 or 1, and call the result H. Then H satisfies* (H) *with Δ, δ, it is (R, ρ)-locally tree-like, $|V(H)| \geq (2\delta/\Delta)n$, and $\Pr(\Phi(G) \text{ SAT}) \leq \Pr(\Phi(H) \text{ SAT})$.*

Proof. Each deleted self-loop lowers the excess by 1, and the later deletions do not lower it (Lemma 2.1), so $\xi(H) \geq \delta n \geq \delta |V(H)|$. Degrees do not grow, and the process stops only when all degrees are at least 2. From $\delta n \leq \xi(H) \leq |E(H)| \leq \frac{\Delta}{2}|V(H)|$ we get $|V(H)| \geq (2\delta/\Delta)n$. Since H is a subgraph of G, distances in H are at least those in G, so the ball of radius R around v in H is a connected subgraph of $B_R(v)$ in G, and it is a tree if $B_R(v)$ is. Hence at most $(2\delta\rho/\Delta)n \leq \rho|V(H)|$ vertices of H have a non-tree ball. Every operation deletes clauses or keeps satisfiability unchanged. □

2.7 The reduction

Theorem 2.15 (Reduction to difficult graphs)**.** *Fix $\varepsilon > 0$, and let Δ, δ be as in Proposition 2.3. For every integer $R \geq 1$ and every $\rho \in (0, 1]$ there is $\beta = \beta(\varepsilon, R, \rho) > 0$ such that for every multigraph G with n vertices and $\xi(G) \geq \varepsilon n$, one of the following holds:*

(a) $\Pr(\Phi(G)\ SAT) \leq e^{-\beta n}$*;*

(b) there is a multigraph H that satisfies (H) *with Δ, δ, is (R, ρ)-locally tree-like, has $|V(H)| \geq \beta n$, and satisfies* $\Pr(\Phi(G)\ SAT) \leq \Pr(\Phi(H)\ SAT)$.

Proof. Let H_1 be the graph of Proposition 2.3, with $n_1 = |V(H_1)| \geq (\varepsilon/\Delta)n$, and set

$$\alpha := \frac{2\delta\rho}{\Delta M_{2R}} \leq \delta.$$

Case 1: H_1 has at least αn_1 pairwise edge-disjoint cycles. By Proposition 2.3, H_1 has maximum degree at most Δ and every component has positive excess, so Corollary 2.12 gives $\Pr(\Phi(H_1)\ \mathrm{SAT}) \leq e^{-\beta_1 n_1}$ with $\beta_1 = \frac{\alpha}{8}\, 4^{-4\Delta/\alpha}$, and (a) holds for every $\beta \leq \beta_1\varepsilon/\Delta$.

Case 2: otherwise. The self-loops of H_1 are pairwise edge-disjoint cycles, so there are fewer than $\alpha n_1 \leq \delta n_1$ of them. If at least $(2\delta\rho/\Delta)n_1$ vertices had a non-tree ball $B_R(v)$, Lemma 2.13 would give at least αn_1 vertex-disjoint, hence edge-disjoint, cycles. So Lemma 2.14 applies to H_1, and the resulting H has $|V(H)| \geq (2\delta/\Delta)n_1 \geq (2\delta\varepsilon/\Delta^2)n$. Then (b) holds for every $\beta \leq 2\delta\varepsilon/\Delta^2$.

Taking $\beta := \min\{\beta_1\varepsilon/\Delta,\ 2\delta\varepsilon/\Delta^2\}$, which depends only on ε, R, ρ, proves the theorem. □

Remark 2.16 (Combination with the paper and order of the constants). The constants are chosen in the following order: ε is given; Proposition 2.3 fixes Δ and δ; Theorem 1.12 of the paper gives $R_0(\Delta, \delta)$ and $\rho_1(\Delta, \delta)$, and we may assume $\rho_1 \leq 1$; Theorem 2.15 is applied with $R = R_0$ and $\rho = \rho_1$, which fixes α and β. In case (b), the graph H is difficult. Since $c(\mathrm{Sol}(\Phi(H))) \geq 1$ whenever $\Phi(H)$ is satisfiable, the main identity (Theorem 1.10 of the paper) and Theorem 1.12 give $\Pr(\Phi(H)\ \mathrm{SAT}) \leq \mathbb{E}\, c(\mathrm{Sol}(\Phi(H))) \leq 2e^{-\gamma|V(H)|/2}$, which is $e^{-\Omega(n)}$ because $|V(H)| \geq \beta n$. Together with case (a) this proves the main theorem. There is no circularity, because R_0 and ρ_1 depend only on Δ and δ.